AKI: KDIGO, biomarkers and care bundle

Definition and stages

- Creatinine ↑ ≥ 0.3 mg/dL (26.5 µmol/L) in 48 h, or ≥ 1.5× baseline in 7 days, or UO < 0.5 mL/kg/h ≥ 6 h
- Stage 1: 1.5–1.9×; stage 2: 2–2.9× or oliguria ≥ 12 h; stage 3: ≥ 3×, Cr ≥ 4 mg/dL, RRT, anuria ≥ 12 h

Causes

- Pre-renal, renal (ATN, AIN, GN), post-renal; sepsis most common in ICU

New biomarkers

- **TIMP-2 × IGFBP7 (NephroCheck) > 0.3**: stress; NGAL, KIM-1, cystatin C, L-FABP
- Detect injury before creatinine rises; guide KDIGO bundle (PrevAKI)

KDIGO bundle

- Stop nephrotoxins, optimise volume and MAP, monitor Cr/UO, avoid hyperglycaemia, avoid contrast

Key point: Biomarkers flag the stressed kidney early: act with the bundle.

Notes: AKI INCLUDE NEW MARKERS p.1

CRRT: principles, dose and anticoagulation

Solute removal

- Diffusion (CVVHD, small molecules), convection (CVVH, middle molecules), combined (CVVHDF)
- Ultrafiltration for fluid removal

Dose

- **Delivered effluent 20–25 mL/kg/h** (prescribe 25–30); RENAL & ATN: higher dose no benefit

Anticoagulation

- **Regional citrate first line** (KDIGO): longer filter life, less bleeding (RICH)
- Citrate toxicity: total/ionised Ca ratio > 2.5; heparin if citrate contraindicated (liver failure)

Prolonging filter life

- Good catheter (right IJ, femoral), pre-dilution, blood flow 150–250, minimise interruptions

Key point: Citrate is the standard: watch the Ca ratio for accumulation.

RRT indications and timing

Absolute indications (AEIOU)

- Refractory acidosis (pH < 7.15), K > 6.5 despite treatment, uraemic complications
- Fluid overload unresponsive to diuretics, dialysable toxins

Timing evidence

- ELAIN: early benefit (single centre); AKIKI, IDEAL-ICU: no benefit early
- **STARRT-AKI: accelerated = standard; more RRT dependence early**
- AKIKI-2: very delayed harmful

Decision

- Watchful waiting if no urgent indication; furosemide stress test (< 200 mL/2 h predicts progression)

Key point: Start for complications, not creatinine alone; avoid too-late.

Notes: AKI INCLUDE NEW MARKERS p.1

Contrast-associated AKI

Definition and risks

- Creatinine rise ≥ 0.3 mg/dL or ≥ 1.5× within 48–72 h after contrast
- Risks: CKD (eGFR < 30), diabetes, hypovolaemia, heart failure, high dose, intra-arterial

Prevention

- **IV isotonic fluids** (saline/bicarbonate) before and after; stop nephrotoxins/metformin
- Lowest dose of iso-/low-osmolar contrast; NAC not effective (PRESERVE)

Pathophysiology

- Renal vasoconstriction (medullary hypoxia), direct tubular toxicity, viscosity, ROS

Key point: Don't withhold needed contrast: hydrate and minimise volume.

Notes: CIN p.1

ATN versus AIN

Comparison

- ATN: ischaemic/toxic (sepsis, aminoglycosides), muddy brown casts
- AIN: drugs (PPI, beta-lactams, NSAIDs), infection; fever, rash, eosinophilia, WBC casts

Urinary indices

- Pre-renal: FeNa < 1%, FeUrea < 35%, urine Na < 20, osmolality > 500
- ATN: FeNa > 2%, urine osmolality < 350; diuretics affect FeNa

Drug-induced AIN

- Stop offending drug; steroids if no improvement in 3–5 days
- Biopsy if unclear

Key point: Fever + rash + eosinophilia after a new drug → think AIN.

Hepatorenal syndrome

HRS-AKI criteria (ICA)

- Cirrhosis + ascites, AKI by ICA criteria, no response after 48 h albumin 1 g/kg
- No shock, nephrotoxins, structural kidney damage

Pathophysiology

- Splanchnic vasodilatation (NO) → ↓ effective volume → RAAS/SNS activation → renal vasoconstriction

Management

- Stop diuretics/NSAIDs, treat precipitant (SBP, bleeding)
- **Terlipressin + albumin** (CONFIRM); noradrenaline in ICU; midodrine + octreotide if neither

Definitive treatment

- Liver transplantation; TIPS selected; RRT as bridge

Key point: Terlipressin + albumin is first-line; transplant is definitive.

Cardiorenal syndrome and diuretic resistance

Classification

- Type 1 acute cardiac → AKI; 2 chronic cardiac → CKD; 3 acute renal → cardiac
- Type 4 chronic renal → cardiac; 5 systemic (sepsis)

Pathophysiology

- Venous congestion ↑ renal venous pressure, low output, neurohormonal activation

Diuretic resistance

- Double loop dose, infusion; sequential nephron blockade (thiazide, acetazolamide: ADVOR)
- Spot urine Na < 50–70 mmol/L at 2 h → escalate
- Ultrafiltration if refractory

Key point: Congestion, not low output, drives most cardiorenal AKI.

Notes: cardio renal syndrome p.1

Rhabdomyolysis

Causes

- Trauma/crush, immobility, seizures, exertion, hyperthermia, drugs (statins), toxins, infections
- Electrolytes: hypoK, hypoP

Pathophysiology

- Myoglobin: tubular obstruction, direct toxicity (haem iron, ROS), vasoconstriction
- Worse in acidic urine and hypovolaemia

Management

- **Aggressive IV fluids** (UO 200–300 mL/h); bicarbonate/mannitol not proven
- Treat hyperkalaemia; RRT for refractory complications

Complications

- Hyperkalaemia, hypocalcaemia (early) → hypercalcaemia (recovery), compartment syndrome, DIC

Key point: CK > 5000 → protect the kidney with early fluids.

Tumour lysis syndrome

Cairo–Bishop definition

- Lab: ≥ 2 of urate, K, phosphate ↑ (or 25% rise) and Ca ↓, within 3 days before to 7 days after therapy
- Clinical: lab TLS + AKI, arrhythmia, seizure or death

Emergency management

- Hydration 3 L/m²/day; **rasburicase** (avoid in G6PD)
- Treat hyperkalaemia; avoid calcium unless symptomatic; early RRT

Prevention

- Allopurinol (low–intermediate risk); rasburicase in high risk

Key point: TLS is preventable: hydrate and lower urate before chemotherapy.

Therapeutic plasma exchange

Principle

- Removes plasma (antibodies, toxins) and replaces with albumin or FFP (TTP)
- 1–1.5 plasma volumes per session

ASFA category I

- TTP, GBS, myasthenic crisis, anti-GBM, CIDP, hyperviscosity, catastrophic APS

Complications

- Hypocalcaemia (citrate), hypotension, bleeding (fibrinogen loss), infection, catheter problems
- Remove highly protein-bound drugs; give doses after session

Key point: TTP is the classic urgent indication: start without delay.

High anion gap metabolic acidosis

Calculations

- AG = Na − (Cl + HCO3), normal 8–12; correct for albumin (+2.5 per 10 g/L fall)
- Osmolar gap = measured − (2Na + glucose + urea), > 10 suggests toxic alcohol
- Delta ratio: < 1 additional NAGMA; > 2 metabolic alkalosis

Causes (GOLD MARK)

- Glycols, oxoproline (paracetamol), L-lactate, D-lactate, methanol, aspirin, renal failure, ketones

Management

- Treat cause; fomepizole/ethanol + dialysis for toxic alcohols
- Bicarbonate only if pH < 7.1–7.2 with AKI

Key point: Calculate the AG and osmolar gap in every metabolic acidosis.

Notes: HGMA Causes And lactic acidosis p.1

Normal anion gap acidosis and RTA

Urinary anion gap

- UAG = Na + K − Cl; negative = GI loss (NH4 excreted); positive = RTA

RTA types

- Type 1 (distal): urine pH > 5.5, low K, stones
- Type 2 (proximal): bicarbonate wasting, Fanconi, low K
- Type 4: hyperkalaemic, hypoaldosteronism (diabetes, drugs)

Management

- Bicarbonate or citrate (type 1, 2); fludrocortisone/K reduction (type 4)
- Avoid excessive saline (hyperchloraemia)

Key point: Negative UAG means GI loss; positive means renal.

Respiratory acidosis and alkalosis

Interpretation

- Acute: HCO3 ↑ 1 per 10 mmHg PaCO2 rise; chronic: ↑ 3.5–4
- Alkalosis: acute HCO3 ↓ 2; chronic ↓ 4–5

Respiratory acidosis management

- Treat cause (COPD, sedation, neuromuscular), NIV, intubation if fails
- Avoid overcorrection (post-hypercapnic alkalosis)

Effects of severe acidosis

- ↓ Contractility, arrhythmias, vasodilation, ↑ ICP, hyperkalaemia, insulin resistance

Key point: Check compensation to classify acute vs chronic.

Hyponatraemia

Approach

- Serum osmolality, urine osmolality, urine Na, volume status
- SIADH: low serum osm, urine osm > 100, urine Na > 30, euvolaemic

Severe symptomatic

- **3% saline 150 mL over 20 min**, repeat ×2–3; target ↑ 4–6 mmol/L
- Max 10 mmol/L in 24 h (8 if high risk)

Osmotic demyelination

- Dysarthria, quadriparesis after 2–6 days; risk: alcohol, malnutrition, hypoK
- Overcorrection: D5W + DDAVP to re-lower

Key point: Treat severe symptoms fast, then limit total correction.

Hypernatraemia

Causes

- Water loss (renal: DI, osmotic diuresis; extra-renal: GI, skin), hypotonic loss
- Sodium gain (hypertonic saline, bicarbonate)

Water deficit

- **Deficit = TBW × (Na/140 − 1)**; TBW = 0.6 × weight (0.5 women/elderly)
- Correct ≤ 10–12 mmol/L/day in chronic; enteral water or D5W

Diabetes insipidus

- Polyuria, dilute urine (< 300 mOsm/kg)
- Central: DDAVP response; nephrogenic: thiazide, NSAID, stop lithium

Key point: Calculate the deficit, add ongoing losses, correct slowly.

Hyperkalaemia

ECG and causes

- Peaked T, flattened P, wide QRS, sine wave, VF
- Causes: AKI, drugs (ACEI, spironolactone), rhabdo, acidosis, TLS

Emergency treatment

- **Calcium gluconate 10% 30 mL** (stabilises myocardium)
- Insulin 10 U + 25 g glucose; salbutamol 10–20 mg nebulised
- Bicarbonate if acidotic; remove K: SZC, patiromer, dialysis

Shifting wears off

- Insulin/salbutamol last 4–6 h → rebound; need removal (dialysis)

Key point: Calcium protects the heart, but only removal fixes the potassium.

Hypokalaemia and periodic paralysis

Causes

- Shift (insulin, beta-agonists, alkalosis, thyrotoxicosis)
- Losses: GI (diarrhoea), renal (diuretics, RTA, hyperaldosteronism), low Mg

Periodic paralysis

- Familial or thyrotoxic; rapid shift after carbohydrate/exercise
- **Small K doses (10 mEq/h)** to avoid rebound; propranolol for thyrotoxic

Replacement

- IV KCl 10–20 mmol/h (central if > 10); correct Mg
- Monitor ECG and levels

Key point: In periodic paralysis, K is shifted not lost: replace gently.

Notes: hypokalemic periodic paralysis p.1

Calcium disorders

Hypercalcaemia

- PTH high: primary/tertiary hyperparathyroidism
- PTH low: malignancy (PTHrP, bone mets), vitamin D, granulomas, thyrotoxicosis

Severe management

- **IV saline 200–300 mL/h**, then calcitonin; zoledronate; denosumab if CKD
- Dialysis if refractory

Hypocalcaemia

- Causes: low Mg, sepsis, citrate, post-parathyroidectomy, pancreatitis
- **Calcium gluconate 10–20 mL 10% IV over 10 min**, then infusion; correct Mg

Key point: Rehydrate first in hypercalcaemia; correct Mg in hypocalcaemia.

Notes: hypercalcemia, hypocalcemia p.1

Metabolic alkalosis

Generation and maintenance

- Generation: H+ loss (vomiting, NGT), diuretics, alkali load
- Maintenance: hypovolaemia, Cl depletion, hypoK, hyperaldosteronism

Urine chloride

- **Cl < 20: chloride responsive** (vomiting, diuretics after)
- Cl > 20: chloride resistant (hyperaldosteronism, Bartter, Gitelman)

Management

- Saline + KCl for Cl-responsive; acetazolamide if fluid overloaded
- HCl infusion rare; treat aldosterone excess

Key point: Most ICU metabolic alkalosis responds to Cl and K.

Systematic ABG and the Stewart approach

Systematic interpretation

- pH → primary disorder → compensation (Winter: $PaCO_2 = 1.5 \times HCO_3 + 8 \pm 2$)
- AG (albumin-corrected) → delta ratio → oxygenation (A–a gradient)

Stewart approach

- Independent variables: $PaCO_2$, SID (Na + K − Cl − lactate), weak acids (albumin, phosphate)
- Low SID → acidosis (hyperchloraemia); low albumin → alkalosis

Value

- Explains saline acidosis and hypoalbuminaemia; SIG reveals unmeasured anions

Key point: Stewart adds insight in ICU acidosis: check SID and albumin.

Notes: abg ,metabolic acidosiscand alkalosis p.1